EVERY NONEMPTY LOCALLY CLOSED SUBSET OF THE SPACE OF A STACK MEETS THE SET OF POINTS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The construction of the reduced induced substack structure on a locally closed subset. Invoked at the start of the counted proof (source lines 3780-3783) as a cross-chapter pointer. Turns the given subset into an algebraic stack whose space is the subset.
- The statement that an immersion is locally of finite type. Invoked inside the counted proof (source lines 3784-3785). Allows points of finite type of the substack to be compared with those of the ambient stack.
- The comparison of points of finite type under a morphism. Invoked inside the counted proof (source lines 3786-3788). Places the points of finite type of the substack among those of the stack inside the subset.
- The description of points of finite type through closed points of affine presentations. Invoked at the end of the counted proof (source lines 3789-3793). Produces a point of finite type on the substack from a closed point of an affine presentation.

Source reference. Original lemma lemma-finite-type-points-morphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3728–3728. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-type-points-morphism, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-identify-finite-type-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3711–3711. Navigation pointer only: the counted text refers to the Stacks item lemma-identify-finite-type-points, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-immersion-locally-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3514–3514. Navigation pointer only: the counted text refers to the Stacks item lemma-immersion-locally-finite-type, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-remark-stack-structure-locally-closed-subset: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3783–3783. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-remark-stack-structure-locally-closed-subset, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. For any locally closed subset $T \subset |\mathcal{X}|$ we have*

$$T \neq \emptyset \Rightarrow T \cap \mathcal{X}_{ft\text{-}pts} \neq \emptyset.$$

In particular, for any closed subset $T \subset |\mathcal{X}|$ we see that $T \cap \mathcal{X}_{ft\text{-}pts}$ is dense in T.

Proof. Let $i : \mathcal{Z} \to \mathcal{X}$ be the reduced induced substack structure on T, see Properties of Stacks, Remark stacks-properties-remark-stack-structure-locally-closed-subset. An immersion is locally of finite type, see Lemma lemma-immersion-locally-finite-type. Hence by Lemma lemma-finite-type-points-morphism we see $\mathcal{Z}_{\text{ft-pts}} \subset \mathcal{X}_{\text{ft-pts}} \cap T$. Finally, any nonempty affine scheme U with a smooth morphism towards $\mathcal{Z}$ has at least one closed point, hence $\mathcal{Z}$ has at least one finite type point by Lemma lemma-identify-finite-type-points. The lemma follows. □